

Array

1. Create an array of 5 integers and print all elements.
2. Find the length of an array.
3. Access an element at a given index.
4. Change an element at a given index.
5. Loop through an array and print each element.
6. Take 5 integers as input from the user and store them in an array.
7. Print the array in reverse order.
8. Find the sum of all elements in an array.
9. Find the average of all elements in an array.
10. Count how many even and odd numbers are in an array.
11. Check if a given number exists in an array.
12. Find the index of the first occurrence of a given number.
13. Find the largest number in an array.
14. Find the smallest number in an array.
15. Find the second largest number in an array.
16. Sort an array in ascending order (use Arrays.sort()).
17. Sort an array in descending order (built-in methods chara).
18. Replace all negative numbers with zero in an array.
19. Count the number of occurrences of a given number in an array.
20. Remove duplicates from an array (hint: ar ekta array baniye okhane diye dite paro).
21. Merge two arrays into one.
22. Copy elements from one array to another.
23. Swap the first and last elements of an array.
24. Compare two arrays and check if they are equal.
25. Find the common elements between two arrays.
26. Find the second smallest number in an array.
27. Count how many numbers are positive, negative, and zero in an array.
28. Find the difference between the largest and smallest elements.
29. Calculate the prefix sum of an array and print it.
30. Check if the array is sorted in ascending order.
31. Check if the array is a palindrome.
32. Find all pairs of elements whose sum is equal to a given number.
33. Find the frequency of each element in an array.
34. Print all unique elements in the array.
35. Count how many elements are greater than the average of the array.
36. Print all elements which are greater than their immediate neighbours.
37. Move all zeros to the end while maintaining the order of other elements.
38. Find the longest increasing subarray.
39. Reverse the array using the two-pointer approach.
40. Create an array of the cumulative product of elements.
41. Find all elements that appear more than once in the array.

42. Create a new array where each element is the sum of itself and the previous element.

43. Check if all elements in the array are distinct.

44. Count the number of distinct elements in the array

45. Split the array into two halves and swap them.

46. Find the equilibrium index in an array (sum of elements before it equals sum after).

47. Calculate the difference between the sum of even and odd elements.

48. Merge two sorted arrays into a new sorted array

49. In the CSE091 course at BRAC University, an instructor stores the marks of all students in an integer array where each element represents a score between 0 and 100. The instructor now wants a programme that can calculate important statistics about the class. Specifically, the program should determine the overall class average, as well as the highest and lowest marks. It should also be able to count how many students passed the course by scoring at least 40 marks. Later, the instructor decides to apply a mark adjustment where each student's score increases by 5, but no score may exceed 100 after this adjustment. For evaluation purposes, the instructor also wants to know the three highest marks in the class, while keeping the original list intact. Finally, if a particular mark is entered, the program should find and display the first and last positions of that mark in the list or indicate that it does not exist.

50. An e-commerce company keeps track of item prices in an integer array. The company uses special values in this array: -1 indicates that an item has been discontinued, and 0 means the item is free of cost. All other positive values represent valid prices of products. The management team now needs a program to clean and analyse this price list. First, discontinued items should be removed so that only available products remain. Next, all free items should be moved to the end of the list while maintaining the order of the other products. Additionally, for all items priced at 1000 or more, a 10% discount should be applied and the discounted prices stored back in the array. After cleaning the data, the program should report the total cost of all items is still available, and also find the median price by arranging the prices in sorted order.

51. A weather research centre collects the daily maximum temperatures of a city over the course of one month and stores them in an array. The researchers now need a programme that can analyse this temperature data. For their study, they want to generate a list of three-day moving averages across the month to understand short-term trends. They are also interested in identifying the longest sequence of consecutive days where the temperature stayed above 35°C, which they define as a heatwave. In addition, they want to know how many days had temperatures strictly greater than the overall monthly average. Finally, due to a reporting adjustment, they require the data to be rotated to the right by a number k (user input), so that the last k days appear at the beginning of the array.

52. A cricket analyst is reviewing the performance of a team in a T20 match. The team's score for each over is stored in an array of 20 integers, where the first element represents runs scored in the first over and so on. The analyst wants to use this data to draw insights. He is interested in calculating the total runs scored during the powerplay (the first six overs). He also wants to identify the best two-over stretch, meaning the two consecutive overs with the highest combined runs, and report which over this stretch begins from. Furthermore, the analyst needs to know how many boundaries were scored during the innings, where boundaries are defined as overs in which the runs scored were exactly 4 or 6. Finally, given a target score entered by the user, the analyst wishes to determine the first over at which the cumulative runs of the innings meet or exceed this target; if the team never reaches the target, the program should indicate this.

[54 - 57 Senario Based Question : Important for Final: Understand how to read a question]

Prepared and Organized by
BADHON NANDI
Student Tutor & FYAT Mentor
Department of Computer Science and Engineering, Brac University

Editing in progress

Email me for any question additions/enquiries: badhon.nandi@g.bracu.ac.bd